

Monero**USD** Protocol

Whitepaper v3 — The Post-Quantum Private Dollar & Ecosystem · Post-Quantum ZK · Block-DAG ·
Constant-Size Verification · Ion Swap PQ-DEX · Usernames · USDm Pay

Launch Edition — 2026

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1. Abstract

MoneroUSD (USDm) is a **post-quantum, fully private digital dollar** and the application ecosystem built around it. It pairs a confidential USD-pegged unit of account with a base layer that is private, post-quantum, sound, and fast *by construction* — every transaction hides its sender, receiver, and amount; every cryptographic primitive resists quantum adversaries; and every node verifies the entire chain from a single constant-size proof rather than replaying history.

Privacy comes from post-quantum Full-Chain Membership Proofs (PQ-FCMP++), stealth addressing, and nullifiers, combined with zero-knowledge confidential amounts — never ring signatures, never a transparent ledger. Scale comes from a block-DAG that produces blocks in parallel and a recursive proof system that compresses the whole chain into a proof small enough to verify on a phone. Around this core sits a complete, post-quantum ecosystem: the **Ion Swap private DEX**, human-readable **usernames**, **USDm Pay** for everyday payments, cross-chain bridges, a desktop and browser wallet, a privacy-preserving explorer, sovereign web hosting, and an on-chain IDE — all governed by a bicameral, threshold-controlled process with no single point of control.

What this document describes. This is the launch specification: every capability below is part of the MoneroUSD ecosystem at 100% completion. It is the picture of what you can do on day one.

2. Design Principles — The Pillars

Every layer of MoneroUSD — protocol and apps alike — is held to the same non-negotiable principles. A feature that cannot satisfy all of them is redesigned, never shipped weakened.

PRIVATE

By default

No sender, receiver, amount, or metadata leak at any layer. Witnesses are bound by commitments; proofs run in a hiding (zero-knowledge) configuration.

POST-QUANTUM

Base layer

Membership, stealth, nullifiers, signatures, and recursive verification all use post-quantum primitives. No classical-only crypto in the core.

SOUND

No inflation

Balance and no-inflation gates are enforced in-circuit. Every mechanism ships with an adversarial test that must reject tampering.

FAST & SCALABLE

≤ 1.8 MB

One constant-size proof verifies the chain. Phone-class memory at any DAG width. Scale by adding nodes, never heavier nodes.

Two further principles are co-equal: **decentralization at every layer** — every authority is permissionless or a t-of-n threshold, never a single operator (not even the founder) — and **liveness by construction** — the chain cannot halt. The block-DAG produces blocks unconditionally; finality is an overlay that can delay confirmation but never stops block flow. Validation gates fail closed on a bad item only, never on the network's progress.

3. Architecture Overview

MoneroUSD is a stack. The base layer settles confidential value; the proof layer compresses it to a constant size; the app layer turns it into a usable, private financial system.

MONEROUSD STACK

APPLICATIONS

Ion Swap DEX

USDm Pay

Username

Bridges

Wallets

Explorer ·
IDE · Hosting

PROOF LAYER — verify once, discard

IVC fold (per-block $O(1)$)

Decider → ≤ 1.8 MB proof

Recursive STARK verify

CONSENSUS — confidential & post-quantum

GHOSTDAG (width 32)

PQ-FCMP++ / stealth / nullifier

ZK confidential amounts

TRANSPORT

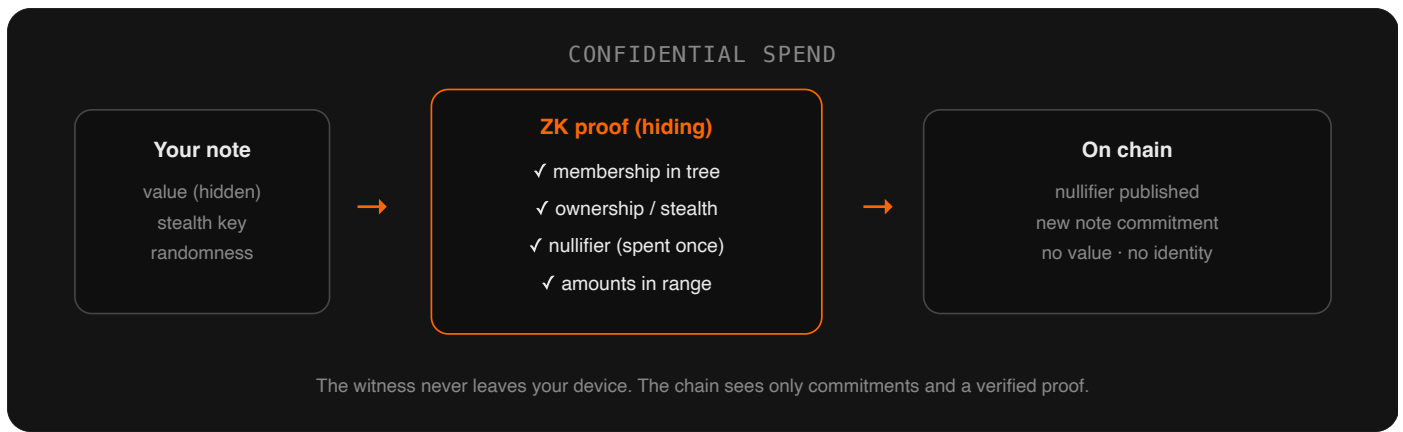
libp2p PQ-gossipsub over Tor v3 onion — onion-only, no clearnet, no raw IP

4. Privacy: PQ-FCMP++, Stealth & Nullifiers

MoneroUSD does not use a transparent ledger and does not use ring signatures. Privacy is built on **PQ-FCMP++** — a post-quantum evolution of Full-Chain Membership Proofs — together with stealth addressing and nullifiers. Every unit of value lives as a hidden *note* committed to the chain. To spend a note, the owner proves three things in zero knowledge, without revealing which note they hold:

- **Membership (PQ-FCMP++)** — the note exists in the global commitment tree, proven by a post-quantum Full-Chain Membership Proof. The anonymity set is *every note ever created* — the entire chain, not a small ring.
- **Ownership & stealth** — the spender controls the note's one-time stealth key, derived so that no two payments to the same recipient are linkable on-chain.
- **Nullifier** — a unique, unlinkable tag that marks the note spent exactly once, preventing double-spends without revealing the note.

Amounts are hidden as commitments and proven in range, so balances are confidential end-to-end. All of this is bound together by Poseidon digests and proven in a hiding (zero-knowledge) configuration: the witness — your keys, your note, your amount — never appears in the public proof.



5. Post-Quantum Cryptography

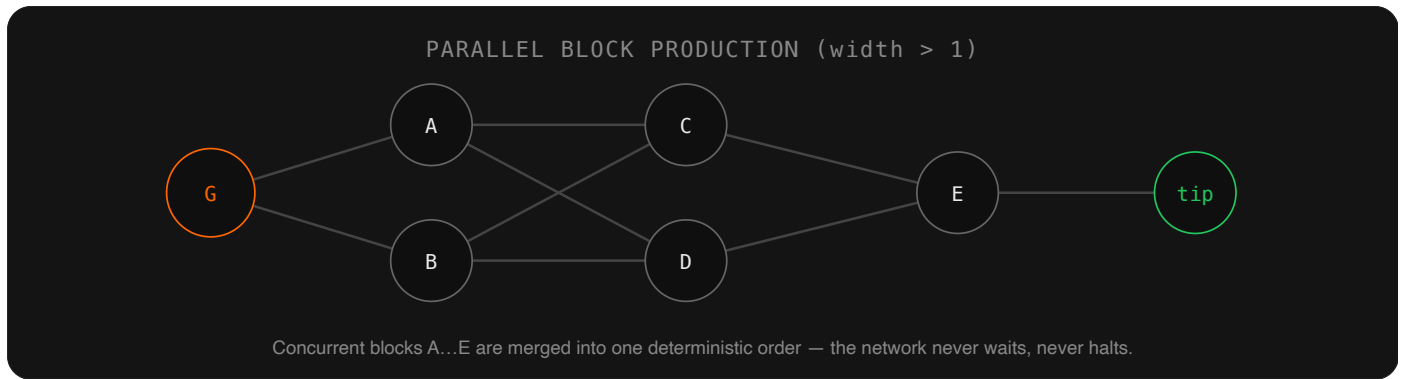
A digital dollar must outlive the arrival of large quantum computers. MoneroUSD therefore uses post-quantum primitives at the base layer — its privacy and authentication do not rest on elliptic-curve assumptions that a quantum adversary could break. Spend authorization, membership, stealth derivation, and nullifiers are all built on hash-based and lattice-grade constructions and proven with STARKs (hash-based, transparent, no trusted setup).

Crucially, verification is **recursive**: a STARK can verify another STARK. That property is what lets the entire chain collapse into one small proof (Section 7) while keeping every guarantee post-quantum. Classical signatures appear only at the edges — for example, when bridging to a non-PQ external chain — never in the MoneroUSD core.

6. The Block-DAG (GHOSTDAG)

MoneroUSD is not a single-file chain. Blocks form a **directed acyclic graph**: many producers can publish at once, each block referencing several parents. A GHOSTDAG ordering (running at width 32, with a one-second target) gives every block a deterministic position in the total order without forcing producers to wait their turn.

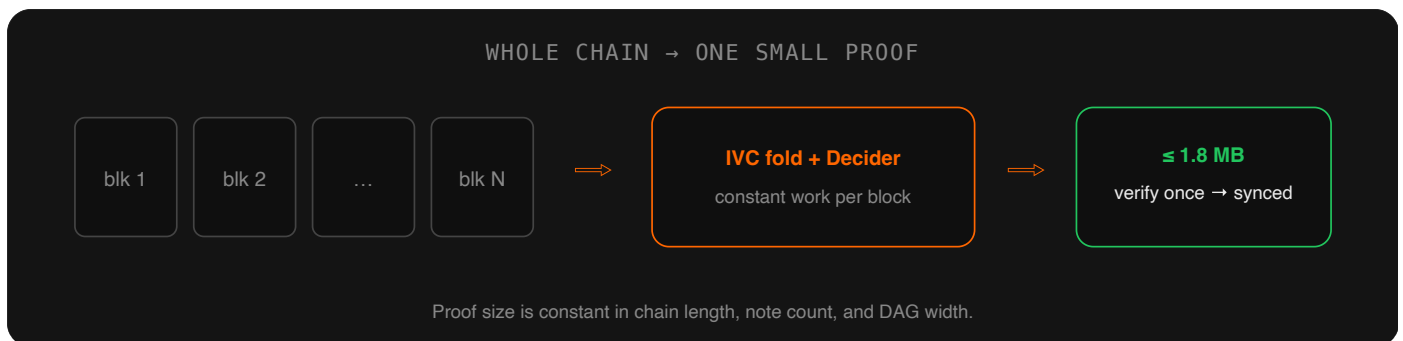
This is the source of **unconditional liveness**. Because blocks are produced in parallel, no single producer — and no committee — can stall the network. There is no "longest chain" race to wedge; there is a mergeset that always advances. Optional carriers such as finality certificates or settlements are never required for a block to be produced.



7. Constant-Size Verification & Instant Sync

Conventional chains force every node to download and replay all history. MoneroUSD does not. Each block's validity is folded into an **incrementally verifiable computation (IVC)**, and a **decider** turns the running fold into a single proof of **constant size (≤ 1.8 MB)** — no matter how long the chain grows, how many notes exist, or how wide the DAG runs.

A new participant **verifies that one proof and is synced**. There is no multi-day catch-up; the proof is verify-once-and-discard. Producing the proof stays within phone-class memory, so the cost of participating does not grow with the network. This is what makes "fast, instant sync" a property of the protocol rather than a promise.



8. Confidential Amounts, Fees & Soundness

Confidentiality is meaningless without soundness: a private chain must still be provably honest. MoneroUSD enforces a **no-inflation invariant in-circuit** — the sum of hidden inputs always equals the sum of hidden outputs plus fees, proven without revealing any of those values. No party, including block producers, can create value out of nothing, and the proof that this holds is part of the same constant-size verification every node already runs.

Fees themselves are confidential. Peer-to-peer transfers are effectively free (a tiny dust floor deters spam, backed by a lightweight client-side proof-of-work stamp rather than a fee market), and protocol revenue is redistributed rather than extracted per transaction. Every new mechanism added to the protocol ships with an adversarial test that *must* reject a tampered input — soundness is verified, not assumed.

9. Finality Gadget

The DAG gives instant, unconditional block flow. A **finality gadget** layers economic finality on top: a threshold of bonded validators co-signs a compact certificate that marks a point in the DAG irreversible. This is strictly an overlay — if validators are offline or slow, blocks keep being produced and confirmed; only the "finalized" marker waits. Finality can delay, but it can never halt. The certificate is itself small and cheap to verify, tracking the tip continuously.

10. L2 Settlement

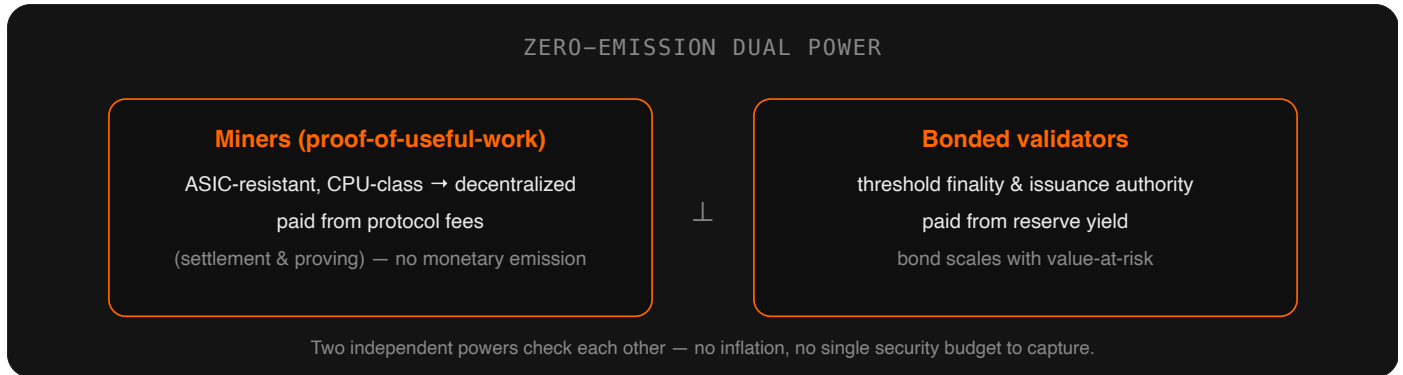
For high-volume application flow, MoneroUSD supports a settlement layer that batches activity and commits it back to the base chain as a single confidential, verified state transition. Applications like the DEX settle through this path, inheriting the base layer's privacy and soundness while keeping per-action cost low. Settlement carriers are optional — they never gate ordinary block production.

11. Onion-Only Transport

Network-level privacy is part of the threat model, not an afterthought. MoneroUSD nodes speak **libp2p PQ-gossipsub exclusively over Tor v3 onion services** — there is no clearnet listener, no raw-IP peering, no legacy block-by-block sync. A node is always addressed by its onion identity, so an observer cannot map the network by IP or correlate a transaction to an origin address. Peer admission is Sybil-resistant by design, and the mixnet-style relay path protects metadata even from the peers that carry it.

12. Nodes & Economics

There are two roles, and both are light. **Nodes** verify the constant-size proof and sample data availability — they do not replay history, so a phone can be a full verifier. **Validators** additionally bond capital to co-sign finality. Security is funded without inflation, through a dual-power model:



Validator bonds are sized to the value the network secures (the cost to corrupt always exceeds the value at risk), and the two powers govern bicamerally (Section 23) so neither can capture the protocol.

13. Decentralized Issuance & The Peg

USDm holds its dollar peg through an **overcollateralized, confidential reserve**. New units are never issued by a single key. Issuance authority is a **post-quantum threshold (FROST-style)** held across independent parties — no operator, and not the founder, can issue unilaterally. Reserve health is maintained adaptively, and reserve yield funds the validator set rather than diluting holders. The result is a digital dollar whose supply is governed by a distributed quorum and whose backing is provable in aggregate while remaining confidential at the level of any individual position.

No inflation, no hidden lever. There is no monetary emission schedule and no single-party control over supply. Issuance is threshold-authorized and bound by in-circuit soundness gates.

14. Wallets

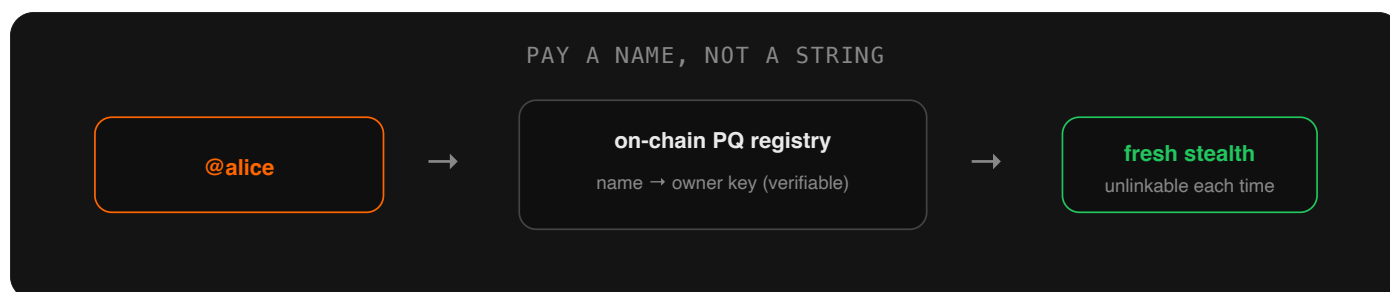
The wallet is where the ecosystem becomes usable. Every MoneroUSD wallet is **local-first**: keys are generated and used on your device and never transmitted. The lineup at launch:

- **Desktop wallet** — full post-quantum send/receive of confidential USDm, a built-in dApp browser, and a local provider that lets web apps request payments and actions without ever exposing your keys.
- **Browser extension & dApp provider** — a companion that injects a `window.moneroUSD` provider so any site can connect, request a confidential payment, or resolve a username, with per-origin stealth addresses for unlinkability.
- **Light wallet + private lookup** — a server-assisted mode that lets thin clients find their notes without revealing which notes are theirs, using a private-information-retrieval side index.
- **Mobile** — native apps performing all cryptography on-device, matching the desktop's local-first guarantees.

15. Usernames — Human-Readable Addresses

Post-quantum stealth addresses are long and unfriendly. MoneroUSD ships a native **username system**: claim a human-readable name once, on-chain, and have it resolve to your wallet — so people can pay *you*, not a wall of characters. Names live in a post-quantum, chain-anchored registry; a claim is a confidential carrier that binds the name to your spend authority, and resolution is verifiable against the same chain everyone else verifies.

- **Own it cryptographically** — a name maps to a key you control; only the owner can update or transfer it, and ownership is provable, not custodial.
- **Privacy preserved** — resolving a name yields a fresh stealth destination, so repeated payments to the same username remain unlinkable on-chain.
- **Ecosystem-wide** — usernames work everywhere: in USDm Pay, in the wallet's send screen, in the dApp provider, and across Ion Swap.



16. USDm Pay

USDm Pay is the everyday payments experience built directly into the desktop wallet, on top of the confidential transfer rail. It turns "send a private dollar" into something as simple as sending a message — and extends to real-world spending through a built-in card — for individuals, online stores, and creators alike.

- **Built-in payment card** — a card integrated into the desktop wallet for spending USDm, so confidential dollars work at checkout and in everyday purchases without leaving your wallet.
- **Pay by username or address** — send confidential USDm to @name or a stealth address; the recipient, sender, and amount stay private by default.
- **Payment requests & links** — generate a request (amount + memo) as a shareable link or QR code; the payer confirms in their wallet and the transfer settles on the PQ rail.
- **Point-of-sale & checkout** — merchants accept USDm through the dApp provider or a hosted checkout, with each request resolving to a fresh stealth destination so a storefront's takings aren't publicly linkable.
- **Confidential by construction** — every payment is a post-quantum confidential transfer; there is no public amount, no public counterparty, and no per-transaction fee extraction.

Because USDm Pay rides the same rail as the rest of the protocol, a payment is final under the same guarantees, verifiable by the same constant-size proof, and private at the same standard as every other action on the network.

17. Ion Swap — Post-Quantum DEX

Ion Swap is a fully post-quantum decentralized exchange. It is not a transparent AMM bolted onto a private coin — value moves through the same post-quantum confidential transfer rail as the rest of MoneroUSD, and trades are authorized by the same zero-knowledge machinery. You can provide liquidity, create tokens, and swap, all without exposing your balances, your counterparties, or your strategy.

RAIL

PQ transfers

Liquidity, swaps, and token actions move via post-quantum confidential carriers — never classical, transparent transfers.

ORDER FLOW

Sealed-bid

Batched, commit-reveal execution makes front-running and MEV cryptographically impossible.

ASSETS

ion1_ tokens

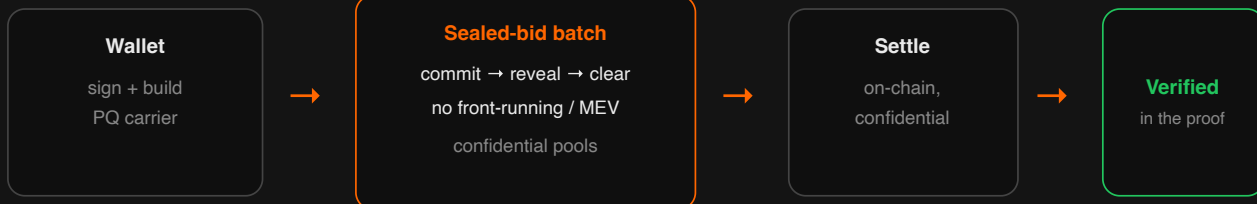
Permissionless custom tokens and pools, with confidential reserves and balances.

CUSTODY

Non-custodial

Keys never leave your wallet; the DEX settles through the chain, not an operator.

ION SWAP — CONFIDENTIAL, POST-QUANTUM



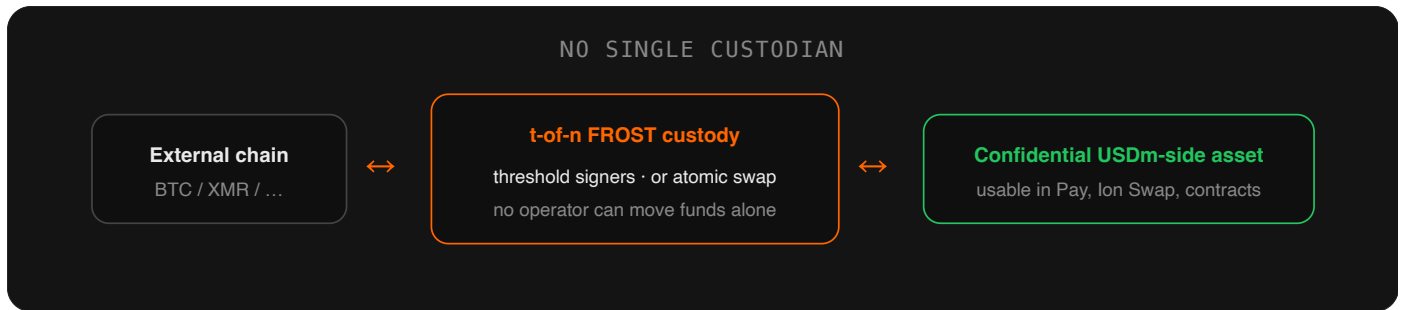
Provide liquidity, create tokens, and swap — balances, counterparties, and order sizes stay private.

18. Custom Tokens & Programmable Privacy

Anyone can create a custom token (ion1_) and liquidity for it, permissionlessly, with confidential supply and balances. Beyond simple tokens, MoneroUSD supports **programmable privacy**: contracts whose state, inputs, and events are hidden by construction. Logic executes deterministically; outputs are committed and proven rather than published; and events are sealed to the recipient's viewing key. Front-running is prevented by commit-reveal, and the execution model is structured so that whole classes of contract exploits are impossible rather than merely discouraged.

19. Cross-Chain Bridges

MoneroUSD connects to the outside world without a custodian. Bridging uses a **threshold (FROST-style) custody quorum** — funds on the external side are controlled by a t-of-n signer set, never a single operator — and, where the counterpart chain allows, **atomic swaps** remove custody from the path entirely. Assets bridged in become confidential MoneroUSD-side representations usable across the ecosystem (in Pay, in Ion Swap, in contracts). Classical signatures used to talk to a non-PQ chain are confined to the bridge endpoint; the MoneroUSD side stays post-quantum.



20. Privacy-Preserving Explorer

A public explorer lets anyone confirm the network is healthy — block flow, the DAG tip, finality progress, and the constant-size proof — **without** exposing what conventional explorers leak. Because amounts, senders, and receivers are confidential at the protocol level, there is nothing to surveil: the explorer shows that the chain is sound and live, not who paid whom.

21. Sovereign Web Hosting

Censorship resistance is incomplete if the front-end can be taken down. MoneroUSD therefore lets its own sites — the main site, Ion Swap, the explorer, dashboards — be served **peer-to-peer from inside the wallet**, with chain-anchored content addressing. If any domain is removed, every user's wallet keeps serving a byte-identical copy. The interface to the ecosystem is as unstoppable as the ledger underneath it.

22. MoneroUSD IDE

To build on confidential infrastructure you need confidential-aware tooling. The MoneroUSD IDE provides an environment to write, test, and deploy programmable-privacy contracts and token logic, with the privacy model and soundness gates built into the developer workflow — so "private by construction" extends from the protocol all the way to the apps people build on it.

23. Bicameral Governance

Upgrades are decided by **two independent houses** — miners and bonded validators — that must agree, so neither stake nor hashpower alone can capture the protocol. Changes are enacted within constitutional bounds by a threshold amendment process, every proposal is publicly auditable, and every enactment is chain-anchored and reversible by the same process. No single party — including the founder — can unilaterally change the rules, move funds, or halt the network.

24. Security & Soundness

MoneroUSD is adversarial by design. Beyond the in-circuit no-inflation and tamper gates, the network assumes a hostile environment at every layer: peer admission resists Sybil attacks; the onion transport resists traffic analysis and flooding; the DAG resists liveness attacks; and bridge custody resists single-signer compromise. Each mechanism carries adversarial tests that must reject the attack, and the trust surface is minimized everywhere — settlement authority is bound to the chain, not to any operator. The guiding standard is simple: ask whether any one party can control, censor, halt, or inflate a layer; if the answer could be yes, it is redesigned until it is no.

25. Comparison

PROPERTY	TRANSPARENT STABLECOINS	PRIVACY COINS	MONEROUSD
Confidential amounts	No	Often	Yes, by default
USD-pegged	Yes	No	Yes

PROPERTY	TRANSPARENT STABLECOINS	PRIVACY COINS	MONEROUSD
Post-quantum	No	Rarely	Yes, base layer
Sync cost	Full history	Full history	One ≤ 1.8 MB proof
Parallel blocks	No	No	Yes (DAG)
Private DEX & payments	No	Limited	Ion Swap + USDm Pay
Single point of control	Issuer	Varies	None (threshold)

26. Launch Readiness

At 100% completion, MoneroUSD is a complete private financial system, not a single coin:

- **Protocol:** post-quantum confidential transfers, GHOSTDAG block production, constant-size verification with instant sync, finality gadget, L2 settlement, onion-only transport, no-inflation soundness.
- **Money:** USDm confidential dollar, threshold-issued, overcollateralized; zero-emission dual-power security.
- **Apps:** Ion Swap post-quantum DEX, USDm Pay, usernames, custom tokens & programmable privacy, cross-chain bridges, privacy-preserving explorer, sovereign web hosting, and the IDE.
- **Clients:** desktop, browser dApp provider, light wallet, and mobile — all local-first.
- **Governance:** bicameral, threshold-controlled, fully auditable.

27. Conclusion

MoneroUSD proves that privacy, stability, post-quantum security, and scale are not a series of trade-offs to be balanced — they can all be true at once, by construction. A confidential dollar you can verify on a phone; a DEX where no one can see your trade; payments as simple as a username; a network no single party can stop. That is the launch vision, and the whole ecosystem is built to deliver it from day one.

The MoneroUSD protocol is open source. Verify everything. Trust nothing.

— *Cas*